

SeedBox

DIMENSIONAL MORPHOMETRY

User Manual

Dimensional seed morphometry

Version 2.01

November 2026

Contents

1. Introduction 4
1.1 About SeedBox 4
1.2 Key features 4
1.3 System requirements 4
2. Installation 5
2.1 Installing SeedBox 5
2.2 Windows security notice 6
2.3 Updating and uninstalling 6
3. Getting to know the interface 7
4. Preparing the data 9
4.1 How to measure the seeds 9
4.2 Spreadsheet layout 9
4.3 Automatic column recognition 9
4.4 File formats 10
5. Entering the data 11
5.1 Importing a spreadsheet 11
5.2 Manual entry 11
5.3 Editing the table 11
5.4 Checking the data 12
5.5 Example dataset 12
6. Running the analysis 13
6.1 Run 13
6.2 Minimum requirements for each analysis 13
6.3 Analysis settings 13
6.4 Changing the data after the analysis 14
7. Results 15
7.1 Overview 15
7.2 Physical properties 15
7.3 Group comparison 16
7.4 Multivariate structure 17
7.5 Classification 18
8. Figures 20
8.1 Gallery 20
8.2 Saving a figure 21

8.3 Exporting all figures (ZIP) . . . 21
9. Report and exports . . . 22
9.1 PDF report . . . 22
9.2 Figure ZIP package . . . 22
9.3 Results workbook (Excel) . . . 22
10. Interpreting the results . . . 24
10.1 Good practice . . . 24
11. Language and preferences . . . 25
12. Troubleshooting . . . 26
13. How to cite . . . 27
Appendix A — Physical properties . . . 28
Appendix B — Statistical procedures . . . 29
Appendix C — Shortcuts and quick commands . . . 30

1. Introduction

1.1 About SeedBox

SeedBox — Dimensional Morphometry is academic software for the dimensional morphometry of seeds. From three linear measurements of each seed — length, width and thickness — it computes the derived physical properties, compares treatments or populations, describes the multivariate structure of the data and uses machine learning to assess how well the seed lots can be told apart.

The methodology follows the reference article by Nobrega et al. (2026), published in *Flora*, which applied this protocol to discriminate *Schizolobium parahyba* populations from the Cerrado and the Atlantic Forest. SeedBox is intended for researchers, lecturers and students in seed science, ecology, restoration and forest breeding.

1.2 Key features

- Import of Excel and CSV spreadsheets with automatic column recognition, or direct data entry.
- Per-seed computation of arithmetic and geometric mean diameter, surface area, volume, aspect ratio and sphericity index.
- Comparison of two or more treatments/populations with automatic selection of the statistical test, effect sizes and grouping letters.
- Correlations, principal component analysis, unsupervised clustering and self-organizing map.
- Eight classifiers evaluated by repeated cross-validation, with identification of the best model and a permutation test.
- Eleven publication-quality figures (up to 1200 dpi), PDF report, ZIP figure package and an Excel workbook with all results.
- Interface in Portuguese and English.

1.3 System requirements

Item	Requirement
Operating system	Windows 10 or Windows 11, 64-bit
Disk space	About 400 MB
Memory	4 GB RAM (8 GB recommended)
Display	Minimum resolution 1280 × 720
Permissions	No administrator rights required
Internet connection	Not required

2. Installation

2.1 Installing SeedBox

SeedBox is distributed as a single installer, **SeedBox_2.01_Setup.exe**. It installs into the user's profile and does not require an administrator password, which makes it easy to use on institutional computers.

1. Double-click **SeedBox_2.01_Setup.exe**.
2. Choose the installer language (Português or English) and click **OK**.
3. On the welcome screen, click **Next**.
4. Confirm the destination folder (default: *%LOCALAPPDATA%\Programs\SeedBox*) and click **Install**.
5. When finished, keep the option to run SeedBox checked and click **Finish**.

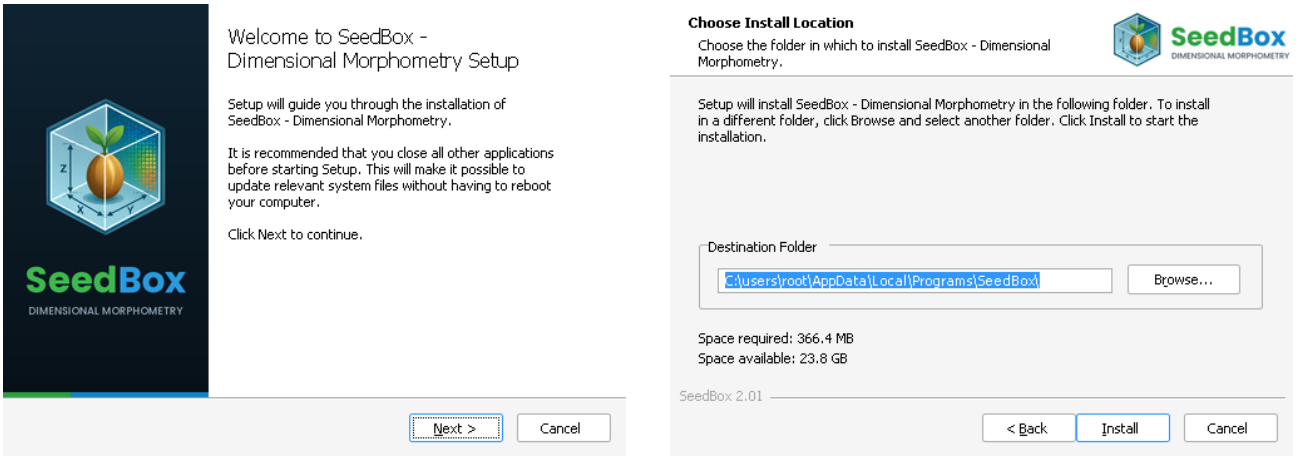


Figure 1. Welcome and destination folder screens.

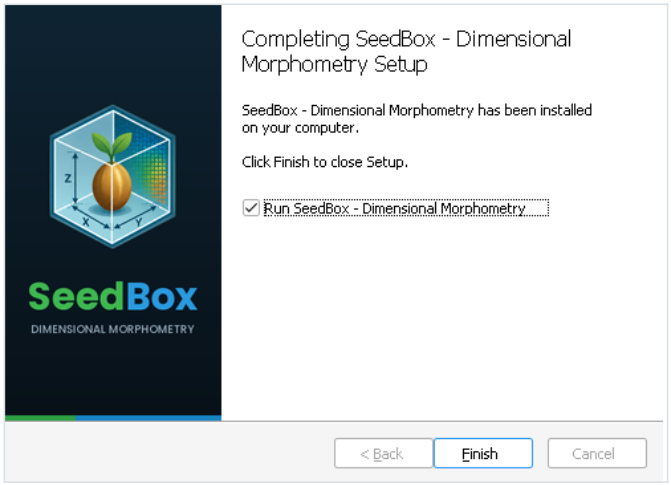


Figure 2. Installation completed.

The installer creates shortcuts on the **Desktop** and in the **Start menu** (SeedBox folder).

2.2 Windows security notice

The first time the installer runs, Windows may display *"Windows protected your PC"* (SmartScreen). This happens with programs that do not carry a commercial digital signature. To continue, click **More info** and then **Run anyway**.

> **Note.** Corporate antivirus software may ask for a similar confirmation. If the installer is blocked, contact your institution's IT support.

2.3 Updating and uninstalling

To update, simply run the installer of the new version: previous files are replaced and user preferences are kept. To uninstall, use Windows **Settings › Apps** or the **Uninstall SeedBox** shortcut in the Start menu.

3. Getting to know the interface

When the program opens, the **Data** page is shown. The window is organized into a side menu, a top bar, the working area of the selected page and a status bar.

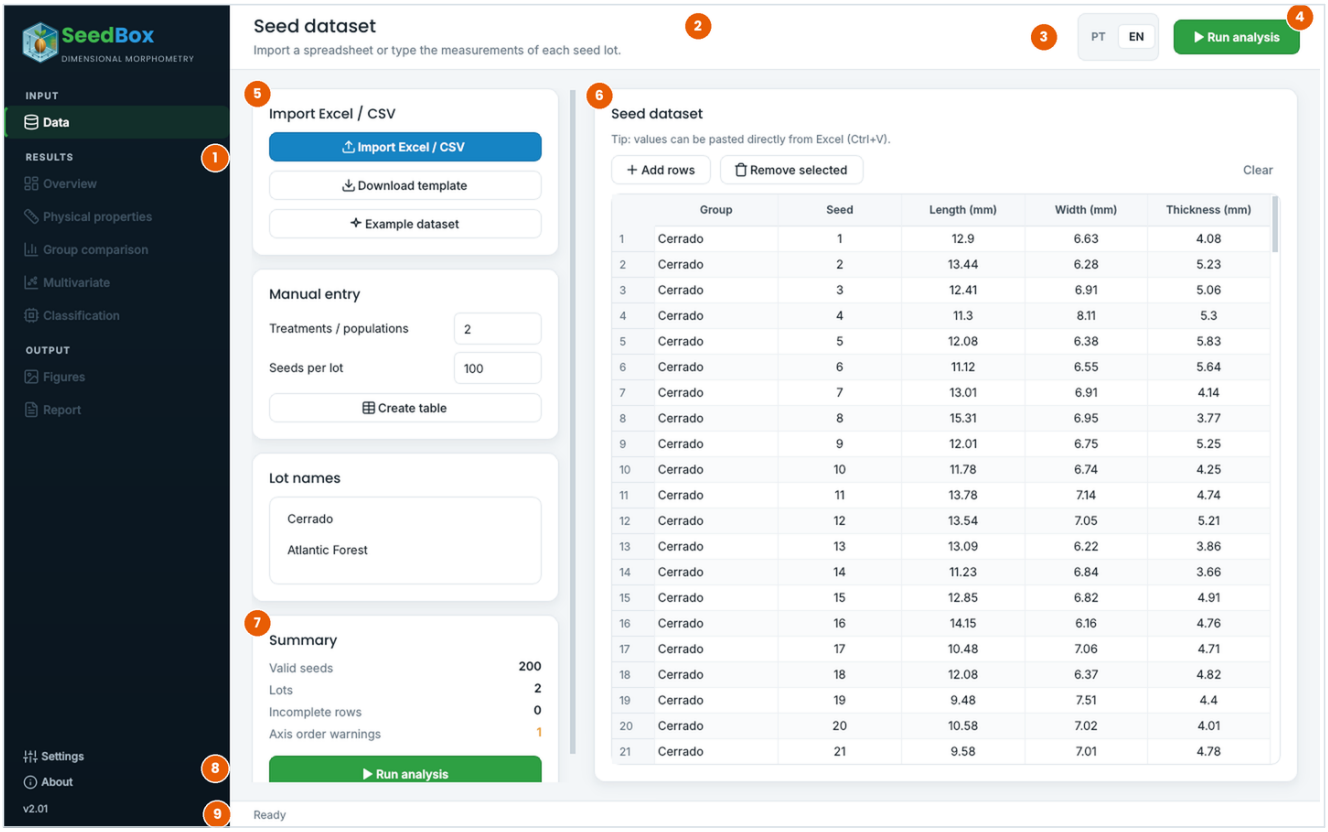


Figure 3. Elements of the main window.

Table 1. Main window elements.

#	Element	Function
1	Side menu	Navigation between pages, grouped into Input, Results and Output. Result pages become available after the first analysis.
2	Page title	Name and short description of the current page.
3	Language	Switches the whole interface, tables, figures and reports between Portuguese (PT) and English (EN).
4	Run analysis	Processes the current data and updates all results.
5	Data input	Spreadsheet import, template, example dataset and manual entry.
6	Data table	One row per seed: lot, seed number and the three measurements.
7	Summary	Count of valid seeds, lots, incomplete rows and warnings.
8	Settings and About	Analysis parameters and program information.
9	Status bar	Progress messages and size of the analysed dataset.

Table 2. Page organization.

Section	Page	Content
Input	Data	Import, entry and checking of data.
Results	Overview	Key indicators, best model and highlights.
	Physical properties	Descriptive statistics by lot and per seed.
	Group comparison	Tests between lots, effect sizes and letters.
	Multivariate	PCA, correlations, clustering and self-organizing map.
	Classification	Classifier performance and best model.
Output	Figures	Figure gallery and export.
	Report	PDF report, figure ZIP and results workbook.

4. Preparing the data

4.1 How to measure the seeds

Each seed should be oriented according to its natural morphology so that the three principal axes are identified consistently:

- **Length (L)** — major axis;
- **Width (W)** — intermediate axis, perpendicular to length;
- **Thickness (T)** — minor axis, perpendicular to the other two.

A digital caliper with 0.01 mm resolution is recommended. All measurements must be in **millimetres**. By definition, L ≥ W ≥ T is expected; the program flags seeds that do not follow this order but does not change the values.

4.2 Spreadsheet layout

Use one row per seed and one column per variable. The first row must contain the column names.

Table 3. Example of an input spreadsheet.

Group	Seed	Length (mm)	Width (mm)	Thickness (mm)
Lot 1	1	12.90	6.63	4.08
Lot 1	2	13.44	6.28	5.23
…	…	…	…	…
Lot 2	1	11.30	8.11	5.30

- The group column identifies the treatment, population or lot. If it is missing, all seeds form a single lot.
- The seed column (number or code) is optional; if missing, numbering is generated automatically.
- Either comma or point may be used as the decimal separator.
- Avoid merged cells, total rows and text below the table.

> **Tip.** Use the **Download template** button on the Data page to obtain a ready-made spreadsheet with the number of lots and seeds set under Manual entry.

4.3 Automatic column recognition

SeedBox identifies columns by their most common names, in Portuguese and English, ignoring accents, case and units in parentheses.

Table 4. Automatically recognized column names.

Variable	Recognized names (examples)
Group	grupo, group, tratamento, treatment, população, population, lote, lot, área, origem, procedência, local, site, matriz, genótipo, acesso
Seed	semente, seed, id, n, n°, número, repetição, replicate, indivíduo
Length	comprimento, comp, length, len, C, L, a, eixo maior, major axis
Width	largura, larg, width, W, b, eixo intermediário, intermediate axis
Thickness	espessura, esp, thickness, thick, E, T, eixo menor, minor axis, altura, height

When the names are not recognized, the program opens the **Column mapping** window (section 5.1).

4.4 File formats

Table 5. Supported file formats.

Format	Notes
Excel (.xlsx, .xlsm)	Recommended. All sheets containing data are read.
CSV / TXT	Semicolon, comma or tab separator, detected automatically.
Excel 97–2003 (.xls)	Open the file in Excel and save it as .xlsx before importing.

Workbooks with several sheets

If each sheet holds one lot (with measurement columns but no group column), each sheet name is automatically used as the lot name. If sheets have different layouts, the mapping window lets you choose the sheet or tick **Use each sheet as a lot**.

5. Entering the data

5.1 Importing a spreadsheet

1. On the **Data** page, click **Import Excel / CSV**.
2. Select the file. If the columns are recognized, the data appear immediately in the table and the status bar reports the number of seeds and lots imported.
3. Otherwise, indicate in the **Column mapping** window which column corresponds to Group, Length, Width and Thickness, check the preview and click **OK**.
4. Check the data in the table and the **Summary** panel.

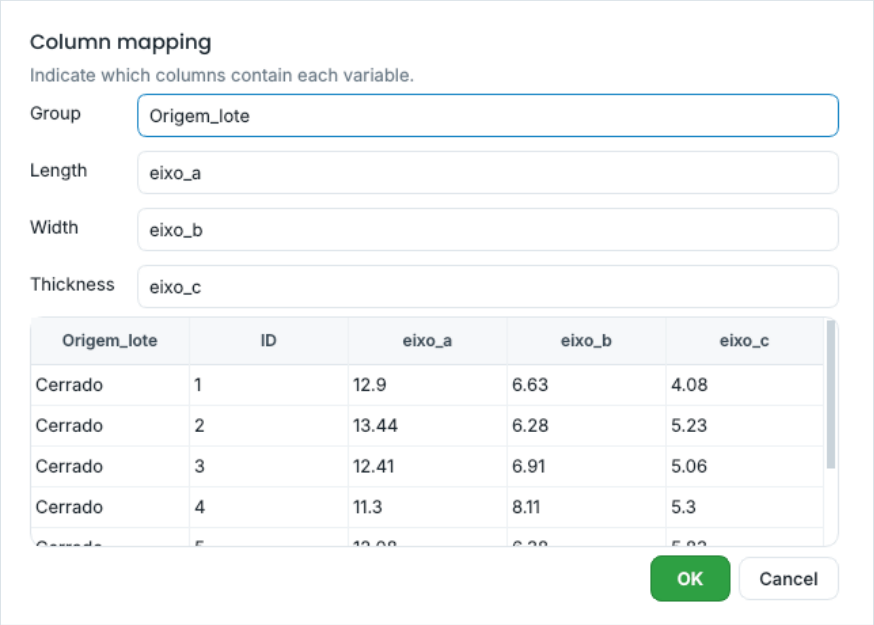


Figure 4. Column mapping window.

> **Note.** Choose “(single lot)” for Group when all seeds belong to the same lot.

5.2 Manual entry

1. Under **Manual entry**, set the number of **Treatments / populations** (1 to 30) and **Seeds per lot**.
2. Click **Create table**. The table is filled with the lots and seed numbers.
3. Rename the lots under **Lot names** (double-click the name). The change is applied to every row of the lot.
4. Type the measurements in the Length, Width and Thickness columns, or paste a block of values copied from Excel (Ctrl+V) starting at the selected cell.

5.3 Editing the table

Table 6. Editing the data table.

Action	How to
Paste from Excel	Select the starting cell and press Ctrl+V. Rows are added when needed.
Copy	Select the cells and press Ctrl+C.
Clear values	Select several cells and press Delete.
Add rows	**Add rows** button (10 rows, in the last lot).
Remove rows	Select the rows and click **Remove selected**.
Clear all	**Clear** button (asks for confirmation).

5.4 Checking the data

The **Summary** panel is updated after every change:

- **Valid seeds** — seeds with all three measurements numeric and positive;
- **Lots** — number of distinct groups;
- **Incomplete rows** — rows with missing, non-numeric or non-positive values; they are excluded from the analysis (invalid cells are highlighted in red);
- **Axis order warnings** — seeds with W > L or T > W; they are analysed normally, but the measurement should be checked.

5.5 Example dataset

The **Example dataset** button loads 200 simulated seeds in two lots (“Cerrado” and “Atlantic Forest”), generated to reproduce the pattern described in the reference article. Use it to explore the program; these values are not real experimental data.

6. Running the analysis

6.1 Run

Click **Run analysis** (top bar or Summary panel). A progress window shows the current step. When it finishes, the program opens the **Overview** page and the status bar reports the total time.

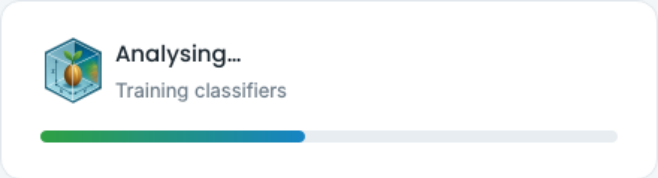


Figure 5. Progress window.

Run time depends on the number of seeds, lots and the computer. For about 200 seeds, the full analysis usually takes 30 seconds to 2 minutes; classifier evaluation is the slowest step.

6.2 Minimum requirements for each analysis

Table 7. Minimum requirements.

Analysis	Requirement
Physical properties and descriptive statistics	≥ 3 valid seeds
Group comparison	≥ 2 lots
Correlations and PCA	≥ 3 seeds per lot
Unsupervised clustering	≥ 6 seeds
Self-organizing map	≥ 10 seeds
Classification	≥ 2 lots, each with ≥ 4 seeds

> **Tip.** For statistically robust results, at least 30 seeds per lot are recommended; the reference article used 100 seeds per population.

6.3 Analysis settings

Under **Settings** (side menu) the parameters below can be adjusted. Choices are saved for future sessions.

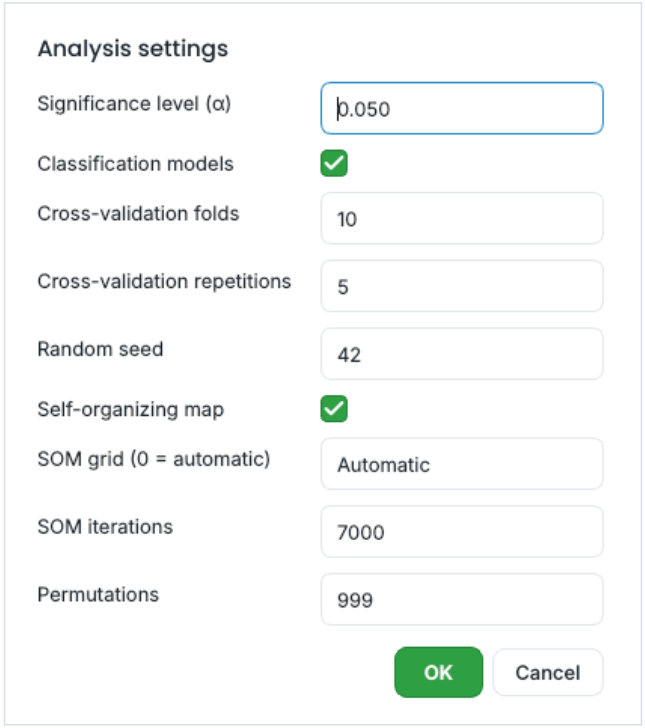


Figure 6. Analysis settings.

Table 8. Analysis parameters.

Parameter	Default	Description
Significance level (α)	0.05	Threshold used in all tests and grouping letters.
Classification models	On	Turns the classification step on or off.
Cross-validation folds	10	Number of folds; reduced automatically when a lot has few seeds.
Cross-validation repetitions	5	Number of cross-validation repetitions.
Random seed	42	Ensures reproducible results.
Self-organizing map	On	Turns the self-organizing map on or off.
SOM grid	Automatic	Hexagonal grid size (side). In automatic mode, the size is chosen from the quantization error.
SOM iterations	7000	Number of training iterations.
Permutations	999	Permutations for the SOM segregation test.

6.4 Changing the data after the analysis

If the data are edited after an analysis, a yellow banner warns that the results shown refer to the previous version. Click **Run analysis** again to update them.

7. Results

7.1 Overview

Shows the main indicators: number of seeds and lots, number of variables with significant differences, variance explained by the first two principal components, the **best** classification **model** with its balanced accuracy, an automatically written list of **highlights** and the mean profile of each lot.

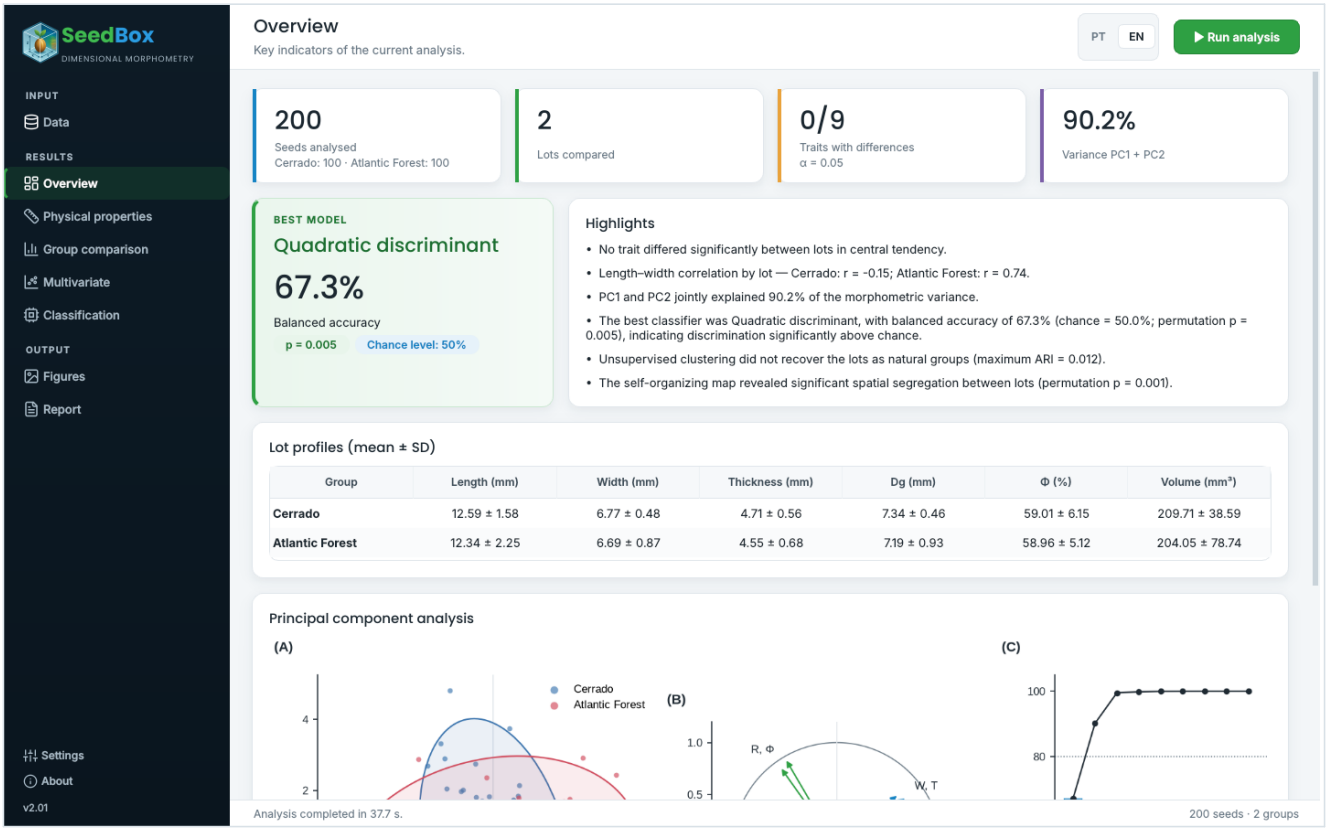


Figure 7. Overview page.

7.2 Physical properties

Organized into four tabs:

- **Summary by lot** — mean ± standard deviation of the nine variables in each lot and pooled;
- **Detailed statistics** — n, mean, SD, standard error, CV, minimum, median, maximum and 95% CI, by lot;
- **Per seed** — values computed for each seed;
- **Assumptions** — Shapiro-Wilk p (normality) by lot and Levene p (homogeneity of variances); values below α are shown in red.

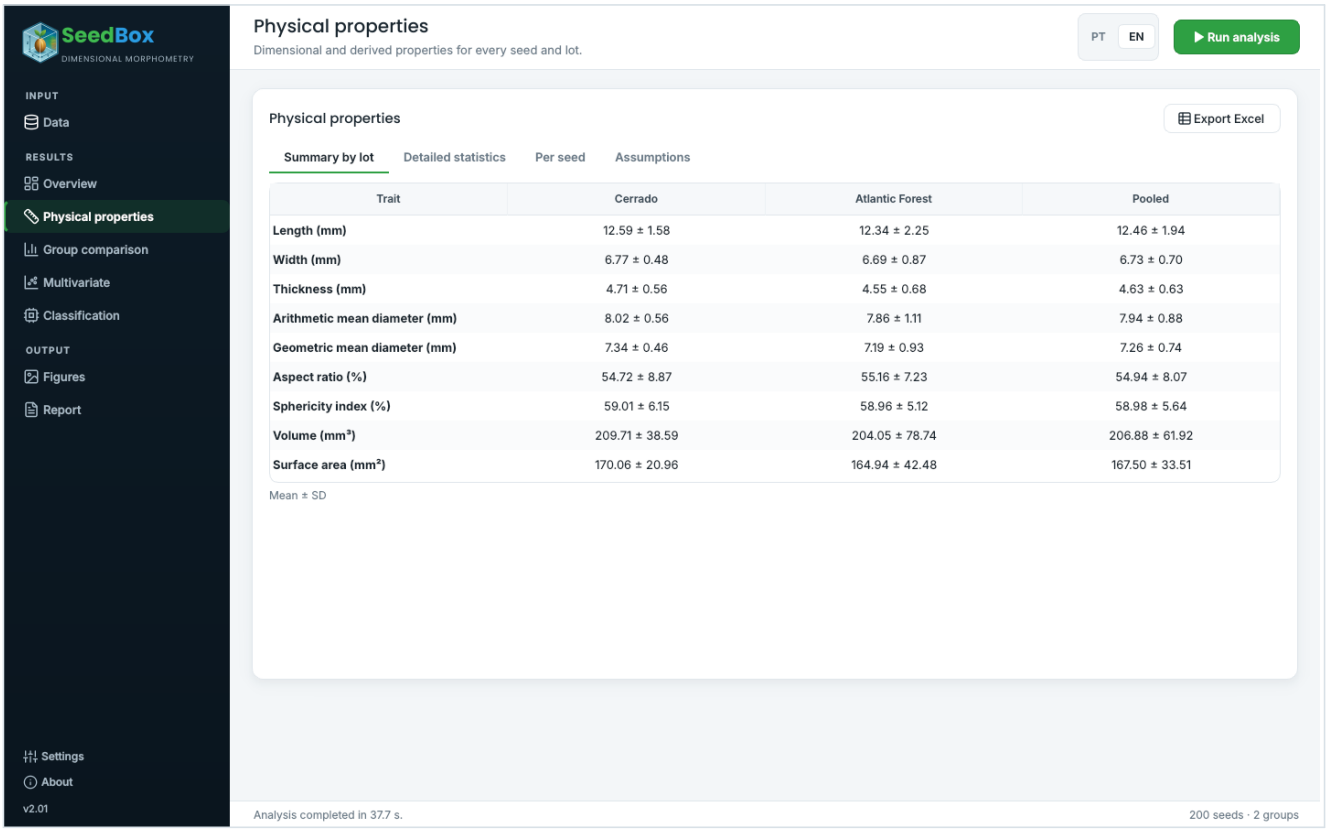


Figure 8. Physical properties page.

The **Export Excel** button saves all numerical results to a workbook (section 9.3).

7.3 Group comparison

For each variable, the table shows each lot's mean ± SD with grouping letters, the test applied, the statistic, the p-value, the effect size and the result. The test is chosen automatically:

Table 9. Choice of test and effect size.

Situation	Two lots	Three or more lots
Normality and homogeneity met	Student t · Hedges g	ANOVA + Tukey HSD · η^2
Assumptions not met	Mann-Whitney U · rank-biserial r	Kruskal-Wallis + Mann-Whitney (Holm) · ε^2

Means followed by the same letter do not differ at level α. With three or more lots, the **Pairwise comparisons** table lists the adjusted p-values of each pair. Below, the distribution figure (violin, box and points) summarizes the differences visually.

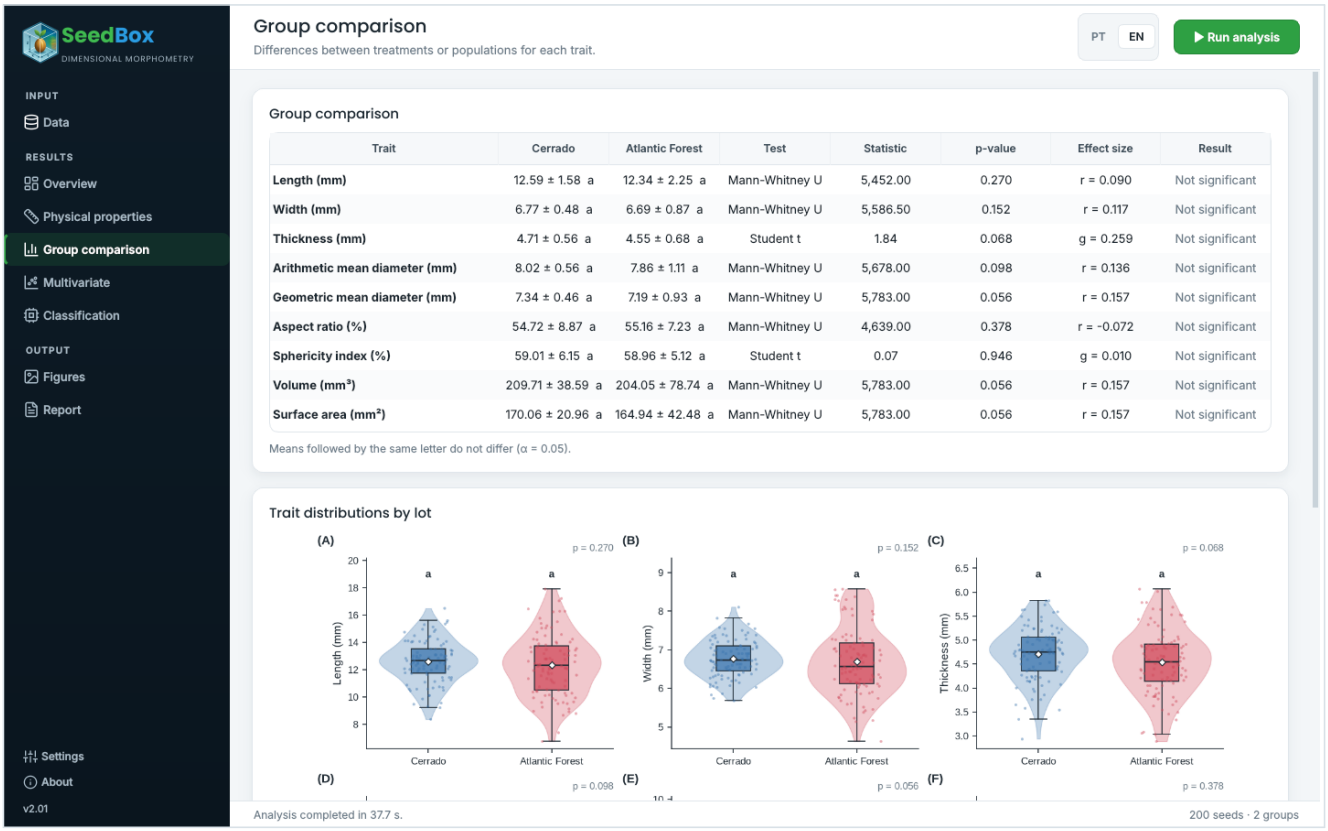


Figure 9. Group comparison page.

7.4 Multivariate structure

- **Principal components** — eigenvalues and explained variance; components with eigenvalue > 1 are highlighted in green.
- **Loadings** — correlation of each variable with the first three components; |loading| ≥ 0.7 is shown in blue.
- **Correlations** — Pearson matrix by lot and pooled; an asterisk marks q < 0.05 after Benjamini-Hochberg correction.
- **Unsupervised clustering** — k-means, hierarchical (Ward) and Gaussian mixture with k equal to the number of lots; ARI measures how well natural groups match the lots, and the k suggested by silhouette and BIC is reported.
- **Self-organizing map** — grid, quantization and topographic errors, χ^2 and permutation p for the segregation of lots on the map.

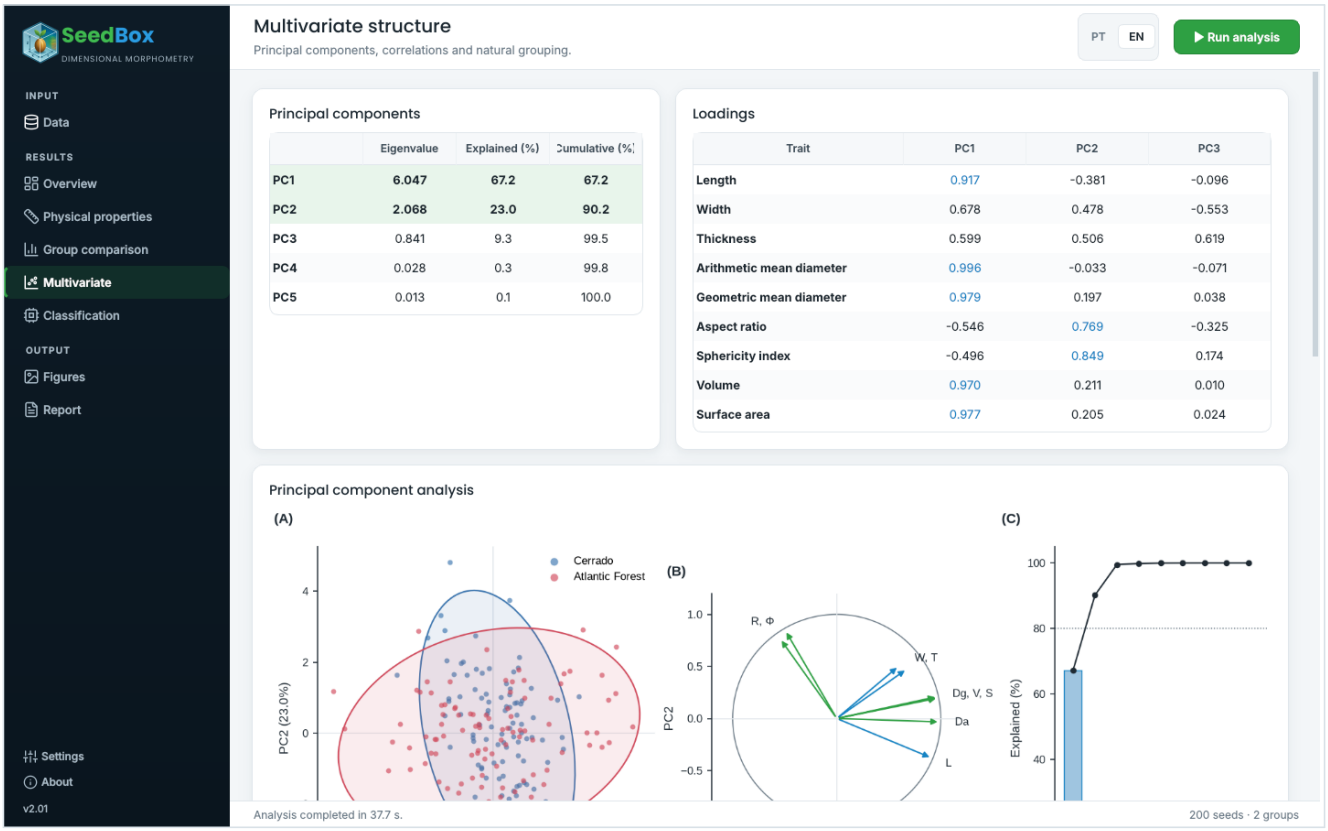


Figure 10. Multivariate page.

7.5 Classification

Eight classifiers are evaluated by repeated stratified cross-validation, always with standardization fitted on the training data of each fold only: logistic regression, linear discriminant, quadratic discriminant, linear SVM, SVM (RBF), k-NN (k = 5), random forest and gradient boosting.

- The **best model** is the one with the highest mean balanced accuracy (ties broken by accuracy).
- The green card shows balanced accuracy, accuracy, F1-score, ROC-AUC, the 95% CI across folds, the validation scheme and the **permutation p**, which tests whether performance exceeds chance.
- The cards on the right compare the best model using all nine variables and only the three measured axes, showing the contribution of the derived properties.
- The table lists all models (mean ± SD) and the figures show performance, the confusion matrix, the ROC curve and the permutation distribution.

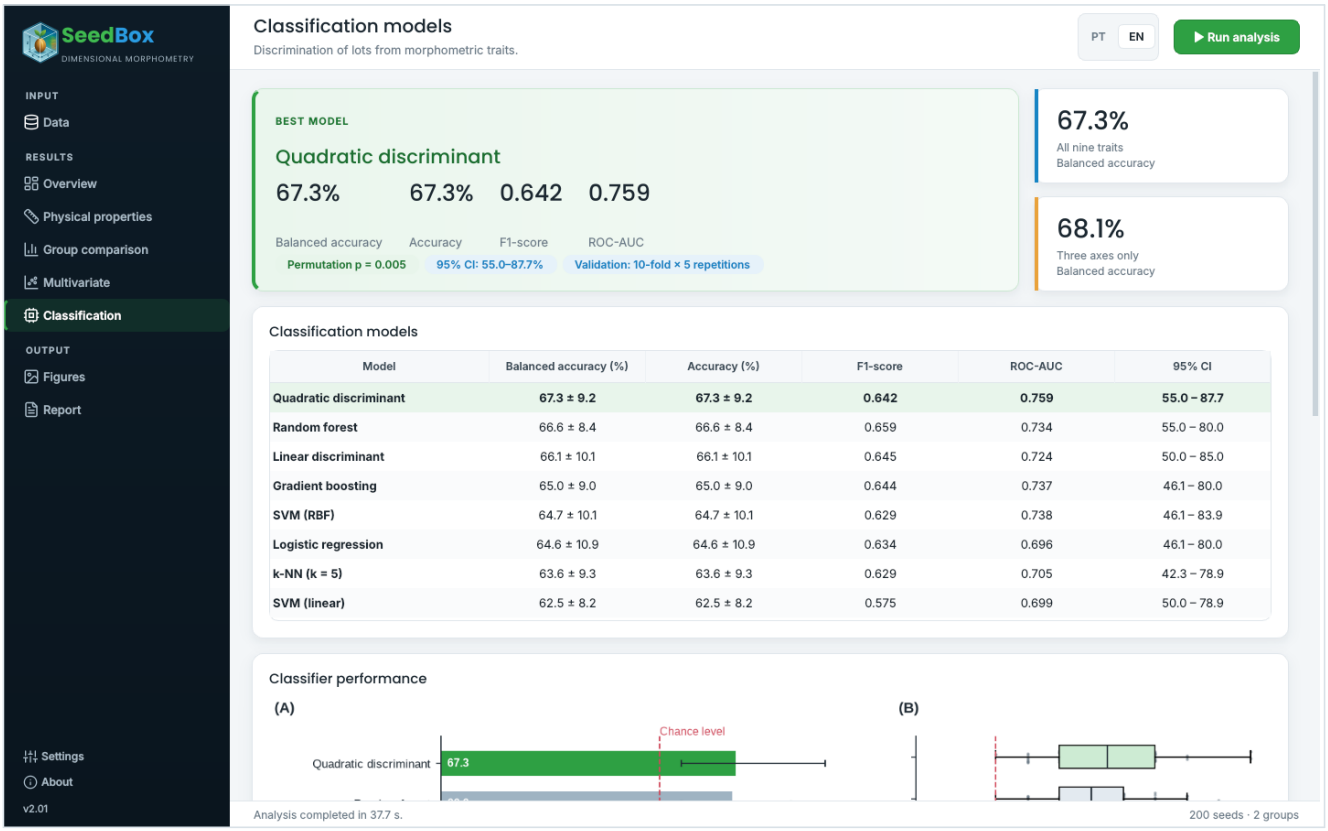


Figure 11. Classification page.

8. Figures

8.1 Gallery

The **Figures** page lists all available figures on the left; selecting one shows its preview on the right. Figures follow the chosen language (including the decimal separator).

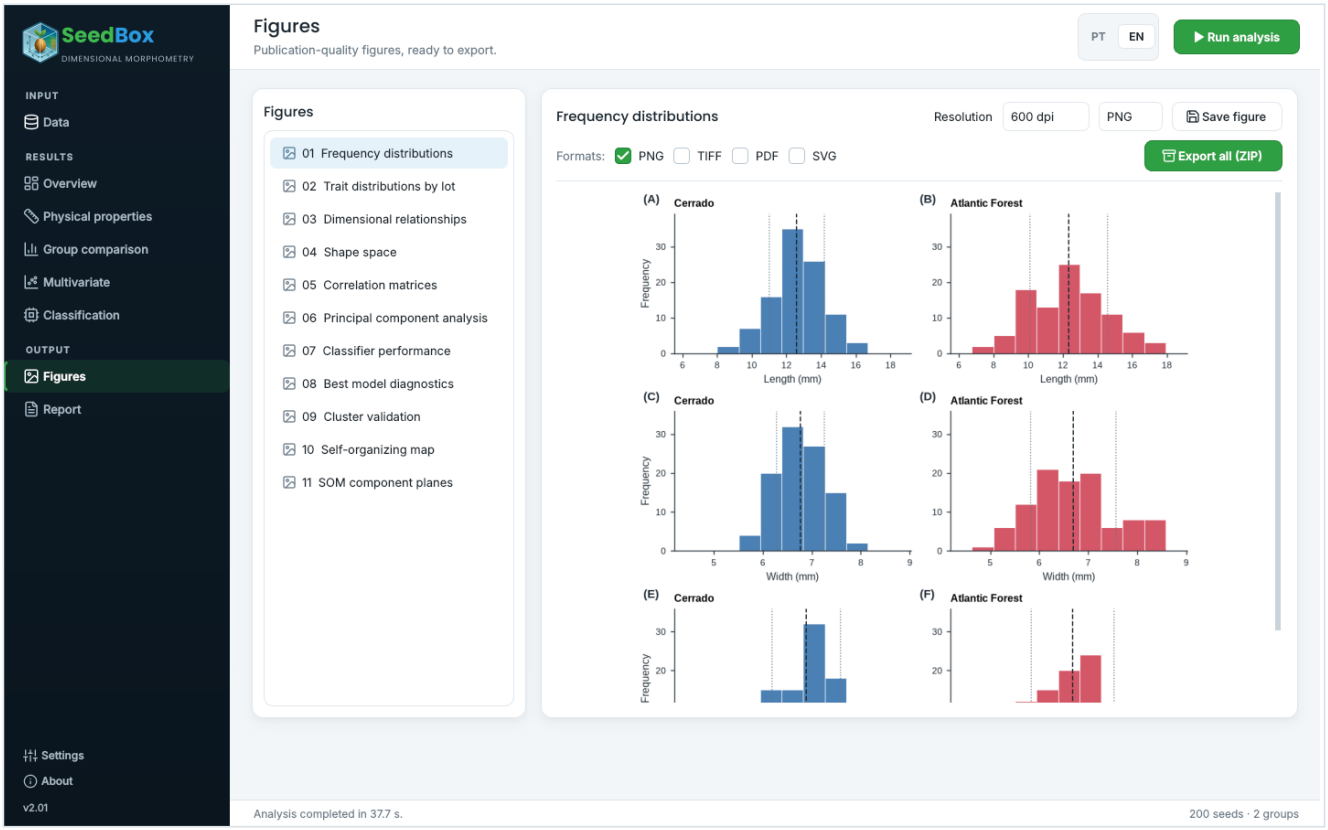


Figure 12. Figure gallery.

Table 10. Figures produced by SeedBox.

#	Figure	Content
01	Frequency distributions	Histograms of length, width and thickness by lot, with mean and ± SD.
02	Trait distributions by lot	Violin, box and points for the nine variables, with letters and p-value.
03	Dimensional relationships	Scatter plots between axes and with geometric diameter, with fitted lines and r by lot.
04	Shape space	Aspect ratio × sphericity and size × sphericity.
05	Correlation matrices	Pearson heatmaps by lot and pooled.
06	Principal component analysis	Scores with 95% ellipses, loading circle and scree plot.
07	Classifier performance	Balanced accuracy with 95% CI and per-fold distribution.
08	Best model diagnostics	Confusion matrix, ROC curve and permutation test.
09	Cluster validation	WSS, silhouette, BIC and ARI.
10	Self-organizing map	U-matrix, dominance map and occupancy by lot.
11	SOM component planes	Distribution of each variable on the map.

Note. Classification, clustering and SOM figures only appear when the minimum requirements in section 6.2 are met.

8.2 Saving a figure

1. Select the figure in the list.
2. Choose the **Resolution** (300, 600 or 1200 dpi) and the format (PNG, TIFF, PDF, SVG or JPG).
3. Click **Save figure** and choose where to save the file.

8.3 Exporting all figures (ZIP)

Tick one or more formats (PNG, TIFF, PDF, SVG) and click **Export all (ZIP)**. The ZIP file contains all numbered figures at the selected resolution.

Tip. TIFF (LZW compression) at 600 dpi meets most journal requirements; PDF and SVG are vector formats, ideal for later editing.

9. Report and exports

9.1 PDF report

1. Open the **Report** page.
2. Optionally fill in Project/experiment, Species, Analyst, Institution/laboratory and Notes.
3. Click **Generate PDF report** and choose where to save it. When finished, the report opens automatically.

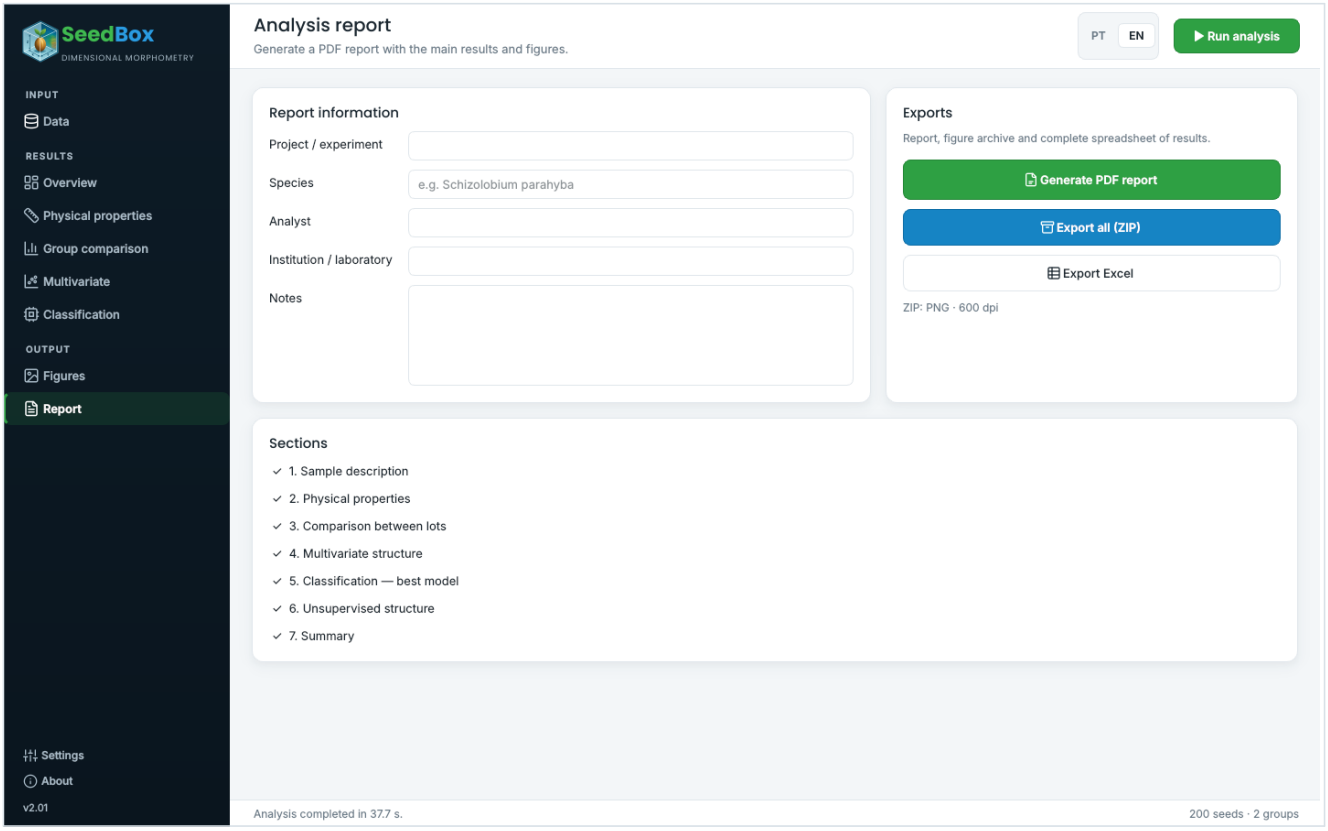


Figure 13. Report page.

The report is produced in the active language and contains: identification and key indicators; sample description; physical properties (mean ± SD with letters); comparison between lots; multivariate structure; best classification model box and table of all models; unsupervised structure; interpretive summary; methods and reference.

9.2 Figure ZIP package

The **Export all (ZIP)** button on the Report page creates a ZIP with all figures as PNG at 600 dpi. For other formats or resolutions, use the Figures page (section 8.3).

9.3 Results workbook (Excel)

The **Export Excel** button creates a workbook with the sheets:

Table 11. Sheets of the results workbook.

Sheet	Content
Per seed	Measurements and properties of each seed.
Summary by lot	Full descriptive statistics.
Assumptions	Shapiro-Wilk by lot and Levene.
Group comparison	Tests, statistics, p, effects and letters.
Pairwise comparisons	When there are three or more lots.
r … / q …	Correlations and q-values by lot and pooled.
PCA / Loadings	Eigenvalues, explained variance and loadings.
Classification	Metrics for all models.
Unsupervised grouping	ARI and silhouette.

10. Interpreting the results

The table below summarizes how to read the main indicators. The thresholds are common references in the literature and should be interpreted in the biological context of each study.

Table 12. Interpretation guide.

Indicator	How to interpret
p-value	$p < \alpha$ indicates a statistically significant difference between lots.
Hedges g	≈ 0.2 small; 0.5 medium; 0.8 large (absolute value).
Rank-biserial r	≈ 0.1 small; 0.3 medium; 0.5 large. Positive when the first lot tends to larger values.
η^2 and ε^2	≈ 0.01 small; 0.06 medium; 0.14 large.
Balanced accuracy	Compare with the chance level (50% for two lots; 1/k for k lots).
Permutation p	$p < \alpha$ indicates that the classifier separates lots above chance.
ROC-AUC	0.5 equals chance; 0.7–0.8 acceptable; 0.8–0.9 good; > 0.9 excellent.
ARI	≈ 0: natural clusters do not match the lots; 1: perfect match.
Silhouette	> 0.5 reasonable to strong structure; 0.26–0.50 weak; ≤ 0.25 no substantial structure.
SOM segregation	$p < \alpha$ indicates that the lots occupy distinct regions of the map.

10.1 Good practice

- Similar means do not imply absence of differentiation: lots may differ in the correlation structure among dimensions, as shown in the reference article.
- A moderate accuracy that is significant in the permutation test indicates a subtle but real morphometric signature; high accuracies with few seeds should be viewed with caution.
- Keep the same random seed to reproduce results exactly, and report the settings used.
- Morphometric differences alone do not distinguish genetic causes, maternal effects or environmental plasticity.

11. Language and preferences

Use the **PT** and **EN** buttons on the top bar to switch the language at any time. Tables, figures, reports and workbooks are then produced in the chosen language, including the decimal separator (comma in Portuguese, point in English). The analysis does not need to be run again.

The language, analysis settings and last folder used are remembered between sessions. The **About** window shows the installed version and the method reference.



Figure 14. About window.

12. Troubleshooting

Table 13. Common problems.

Situation	What to do
“The file could not be read.”	Check that the file is .xlsx, .xlsm, .csv or .txt and that the first row contains the column names. .xls files must be saved as .xlsx.
The mapping window opened.	Column names were not recognized: assign each variable manually.
Incomplete rows or red cells.	There are missing, non-numeric or ≤ 0 values. Correct them; otherwise these seeds will be excluded.
“At least 3 complete seeds are required.”	Enter more seeds with all three measurements.
The Classification page is empty.	At least two lots with four or more seeds each are required, and the option must be on in Settings.
The analysis is slow.	Wait for the progress window. To speed it up, reduce cross-validation repetitions or turn off the SOM.
Results did not change after editing the data.	Click **Run analysis** again (a yellow warning banner is shown).
The report did not open automatically.	Open the PDF from the folder where it was saved; a PDF reader must be installed.
Windows blocked the installer.	Click **More info** › **Run anyway** (section 2.2).
The program closed unexpectedly.	Reopen SeedBox. A technical log is written to *%APPDATA%\SeedBox\seedbox.log*; send it to support if the problem persists.

13. How to cite

When using SeedBox in academic work, please cite the method reference article:

> Nobrega, M.A.S.; Nabuco, M.L.H.P.; Araújo, D.; Fernandes, S.Y.; Miguel, E.C.; Arruda, G.J.; Santiago, E.F.; Santos, J.S.; Pontes, M.S. (2026). A simple method for discriminating *Schizolobium parahyba* (Vell.) Blake (Fabaceae) seeds population by dimensional morphometry and machine learning algorithms. *Flora*, 153108. https://doi.org/10.1016/j.flora.2026.153108

And state the program version used, for example: *"Analyses were performed with SeedBox — Dimensional Morphometry, version 2.01."*

Appendix A — Physical properties

Each seed is treated as an ellipsoid with axes L (length), W (width) and T (thickness), in millimetres. The derived properties are:

Arithmetic mean diameter (mm)

$$D_a = (L + W + T) / 3 \quad \text{(A1)}$$

Geometric mean diameter (mm)

$$D_g = (L \times W \times T)^{1/3} \quad \text{(A2)}$$

Aspect ratio (%)

$$R = (W / L) \times 100 \quad \text{(A3)}$$

Sphericity index (%)

$$\Phi = (D_g / L) \times 100 \quad \text{(A4)}$$

Volume (mm³)

$$V = \pi \times L \times W \times T / 6 \quad \text{(A5)}$$

Surface area (mm²)

$$S = \pi \times D_g^2 \quad \text{(A6)}$$

> **Note.** The surface area corresponds to that of a sphere of diameter D_g, an approximation widely used in seed physical-property studies and adopted in the reference article.

Appendix B — Statistical procedures

- **Descriptive statistics**: n, mean, standard deviation, standard error, coefficient of variation, minimum, quartiles, median, maximum and 95% confidence interval (Student t).
- **Assumptions**: Shapiro-Wilk within each lot and median-centred Levene across lots.
- **Comparisons**: parametric test when all lots meet normality and variances are homogeneous; otherwise a non-parametric test (section 7.3). Letters obtained by the compact letter display from the pairwise comparisons.
- **Correlations**: Pearson, with Benjamini-Hochberg false discovery rate control (q).
- **PCA**: correlation matrix of the nine standardized variables; retention by the Kaiser criterion (eigenvalue > 1).
- **Classification**: repeated stratified cross-validation (default 10 × 5) with standardization inside each training fold; metrics: accuracy, balanced accuracy, macro F1 and ROC-AUC; 95% CI from fold percentiles; label permutation test for the best model.
- **Clustering**: k-means, Ward hierarchical clustering and Gaussian mixture on standardized variables; agreement with lots by the adjusted Rand index; k from 2 to 8 assessed by WSS, silhouette and BIC.
- **Self-organizing map**: hexagonal grid trained with a Gaussian neighbourhood and decreasing learning rate; quantization and topographic errors; segregation of lots assessed by the χ^2 of the neuron × lot table against the distribution obtained by label permutation.

Appendix C — Shortcuts and quick commands

Command	Function
Ctrl + V	Paste values from Excel into the data table.
Ctrl + C	Copy the selected cells.
Delete	Clear the content of the selected cells.
Double-click a lot name	Rename the lot in every row.
Click a row header	Select the entire row.
PT / EN	Switch the language of the interface and outputs.